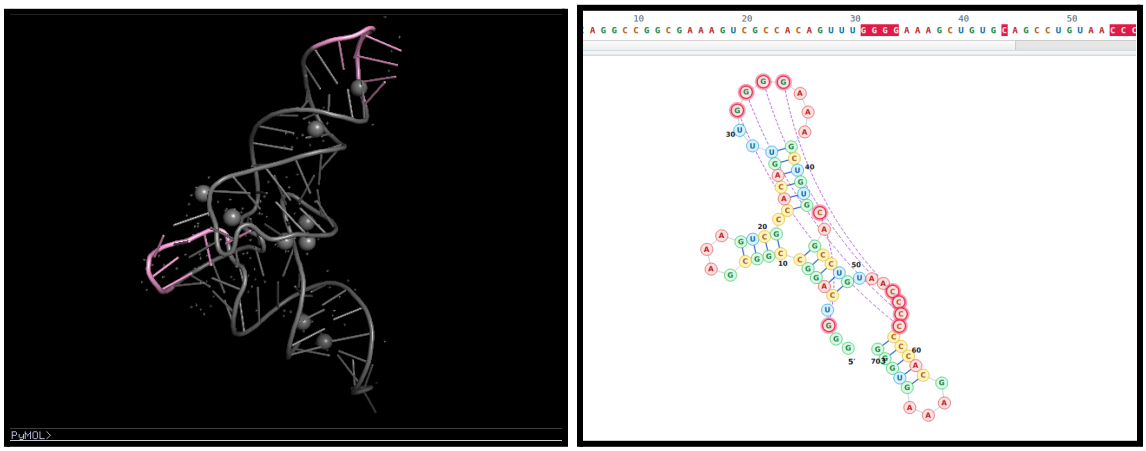
User Manual for Dssr_select v2.0.0 (10/9/26)



Software to accurately annotate and represent motifs and structural features within nucleic acids including different types of pairs, stems, coaxial stacks, G-quadruplexes, pseudoknots, u-turns, and more. It allows users to create visual selections through both PyMOL's molecular viewer and on a built-in 2D viewer with two interfaces: the PyMOL command line and the GUI (Graphical User Interface). It also allows for cartoon-block schematics to show further insights about interactions within DNA structures such as planarity.

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Information Regarding Dssr_select as a software:

- Dssr_select is accessible from its Github page. It is updated constantly and different versions are released and exist in archives or different branches. The following versions of Dssr_select have been released:
    - (Latest): DSSR-PyMOL v2.0.0: Interactive RNA 2D Layout Studio & Bi-directional 3D Integration
    - DSSR-PyMOL v1.1.0
    - DSSR-PyMOL v1.0.1
    - DSSR-PyMOL v1.0.0
- The manual itself is continuously revised and modified in response to feedback from users and the further changes and developments of Dssr_select.

- Dssr_select is fully intended and designed to work as demonstrated in the manual. Any discrepancies that exist will be immediately addressed

- Issues related to the manual or the software can be reported on the Github: https://github.com/xiang-jun/dssr-pymol

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Introduction, Download and Installation (1/2):

Dssr_select is a PyMOL plugin that is able to bridge DSSR (Dissecting the Spatial Structure of RNA) with PyMOL's environment and visualization capabilities. It gives users many unique and novel features:

1. Selecting structural RNA/DNA features: On a loaded structure in PyMOL, users are able to select from up to 17 supported features including things like base pairs, hairpins, stems, junctions, u-turns. G-quadruplexes, pseudoknots, and more such that they are highlighted by PyMOL in its viewer system.
2. Create cartoon block schematics of nucleic acid structures
3. Extract information for sequences on any loaded structure
4. Interact with a 2D secondary structure diagram that mirrors selections with the 3D PyMOL viewer

In order to use Dssr_select, the following prerequisites must be met:

- ☐ PyMOL (with Qt GUI, a standard in the vast majority of modern PyMOL versions)
- ☐ x3dna-dssr DSSR (Dissecting the Spatial Structure of RNA) executable installed such that it is accessible the system PATH and reachable

What's new in v2.0.0 (according to the Github)

1. Interactive 2D RNA Layout Studio (dssr_2d):
    a. Pure-Python 2D Engine: Integration of a lightweight, self-contained secondary structure layout engine ported to Python from ViennaRNA/fornac, requiring no external dependencies beyond PyMOL and DSSR
    b. Multiple Layout Algorithms: Switch between Standard, Circular, Linear, and Radiate Arrangements
    c. Specialized tRNA Cloverleaf Recognition: Includes topology heuristics that automatically identify classical tRNA multi-branch junctions and orient the cloverleaf with the acceptor stem directed downward and the anticodon arm upward

    d. Tertiary & Non-Canonical Contact Overlay: Toggle DSSR-detected non-canonical (non-WC/Wobble) base pairs as dashed arcs to inspect long-range tertiary folding alongside primary stems
2. Bi-directional 2D ⇔ 3D Selection Linking
    a. Interactive 2D-to-#D picking: Click individual bases, drag rectangular selection boxes on the canvas, or use the Brush tool (B) to highlight residues in PyMOL's 3D viewport in real time.
    b. Synchronized 3D-to-2D Mirroring: Selections made in PyMOL's 3D view are reflected back onto the 2D layout canvas and sequence ruler
    c. Interactive Kinematic Editing: Refine 2D diagram layouts with soft drag, base-pair, loop, stem, and whole-branch manipulation, backed by a full undo/redo stack (Ctrl+z / Ctrl+Y).
3. 1D Sequence Ruler with Native PDB Numbering
    a. Displays the complete sequence broken by chains, aligned directly above an authentic residue numbering ruler
    b. Supports click-and-drag range selection, shift-extending, and Ctrl-toggling directly linked to 3D and 2D highlights
4. GUI & Visualization Refinements:
    a. Signature PyMOL Color Palette: Uses PyMOL's signature hot-pink selection accents and classical base-specific colors
    b. Dark Mode Toggle: Added a 1-click Dark button switching the entire GUI, sequence ruler, and 2D canvas between light and PyMOL-inspired slate-dark themes
    c. Smart Filter & Clear: Equipped the feature search box with an embedded clear button and automatic clearing when switching between motif types
    d. Non-Destructive Navigation: Improved key event filtering so pressing Esc clears active highlights across PyMOL, the sidebar, and the 2D nodes without closing the window
    e. Default Whole-Molecule Blocks: The Make blocks button now defaults to generating schematic blocks for the entire loaded structure when no sub-motifs are selected
5. Publication-Ready Diagram Export
    a. Vector & Raster Formats: Export customized 2D RNA diagrams directly to vector SVG or high-resolution PNG
    b. Layout Persistence: Save and reload edited 2D coordinates using JSON or export dot-bracket secondary structure notation directly to the clipboard.

Dssr_select from the CLI (Command Line Interface):

One of the key ways to use Dssr_select is from the CLI (Command Line Interface). The Command Line Interface in PyMOL through a few keys allows users to generate highly aesthetic and very informative visualizations in PyMOL. It also allows for creation of the cartoon block-like schematics that are effective at showing planarity embedded within DNA structural components and features.

The primary command used from the CLI is ‘dssr_select’ which takes a total of 10 parameters:

- selection: determines which atoms in PyMOL to analyze
    - default: selection = all
- state: which state to use when running the program
    - default: state = -1 (to indicate the current state)
- feature: the type of nucleic acid feature/element/motif to be selected
    - default: feature = pairs
- index: the instance of the feature
    - default: index=1 (1 means the first instance)
- name: the name of the selection object that is created from calling the command
    - default: name = “dssr_select”
- exe: the path to the DSSR executable
    - default: exe = “x3dna-dssr”
- show_info: whether or not to print extra DSSR output to the console
    - default: show_info = 0 (tells DSSR not to print extra output)
- quiet: controls if the command being called will print a message
    - default: quiet = 1 (meaning no message is printed)
- color: the color that is applied to the selected feature(s)
    - default: color = “auto” (by default, most features get pink)
- precolor: determines if the entire structure is initially set to gray before a feature is selected to show contrast

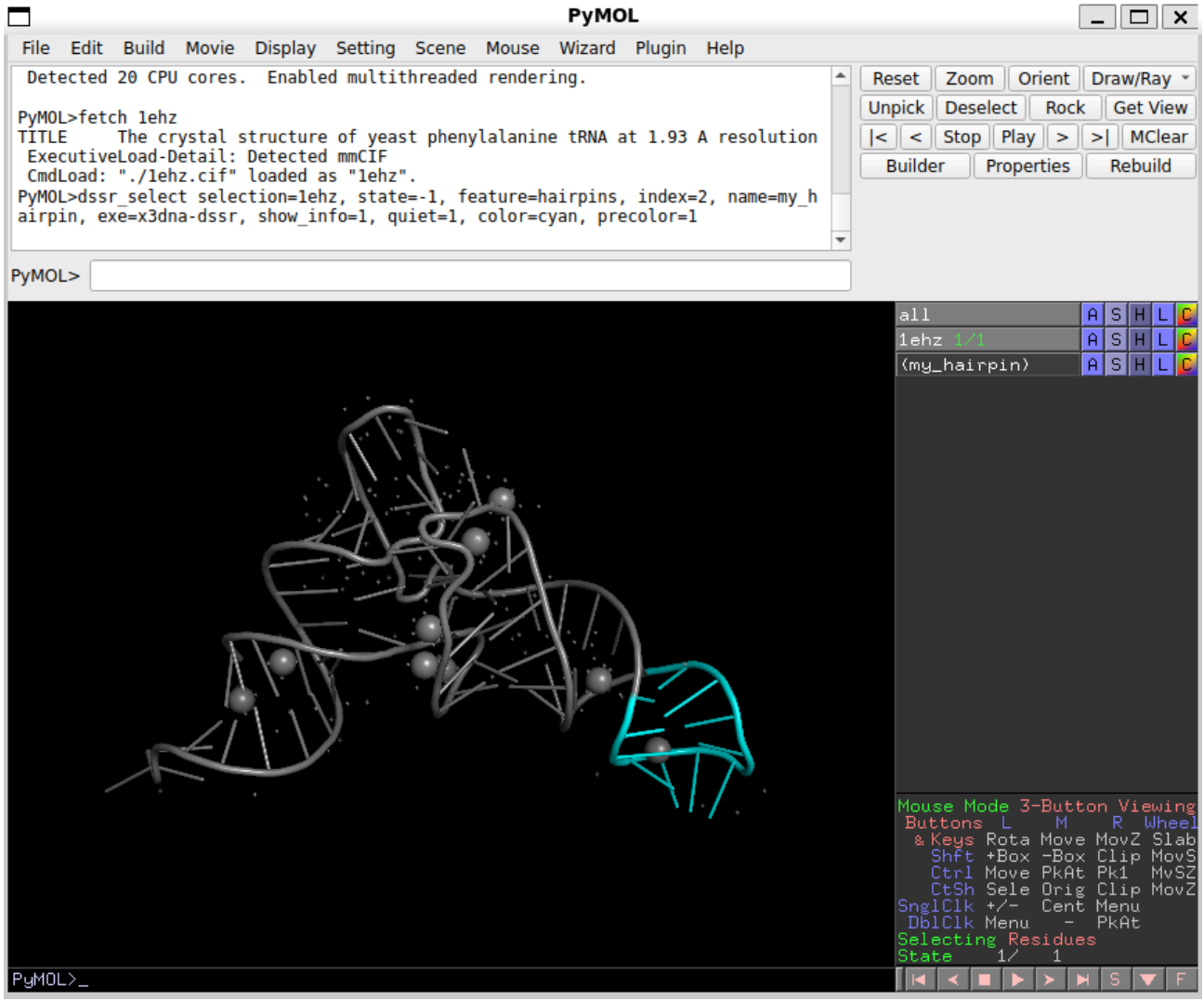
- default: precolor = 1 (sets the entire structure to gray)

Sample One: To use all possible parameters, run the following lines from the CLI in PyMOL

1. fetch 1ehz
2. dssr_select selection=1ehz, state=-1, feature=hairpins, index=2, name=my_hairpin, exe=x3dna-dssr, show_info=1, quiet=1, color=cyan, precolor=1

Breakdown: the 'fetch command' tells PyMOL to bring up the yeast crystalline structure (1ehz). Then, the dssr_select command is called, the selection is set to 1ehz meaning those are the atoms in PyMOL to be analyzed, the state is set to -1 to be current, the feature being selected is a hairpin, it is the second instance of a hairpin, the selected object will be named "my_hairpin", extra DSSR output will not be printed because show_info is set to 0, the command itself is set to 1 so it will not print a message, the color is cyan so the selected feature will be shown in light blue, and precolor is set to 1 so the rest of the structure will be colored in gray.

The following image is produced by PyMOL's molecular viewer:



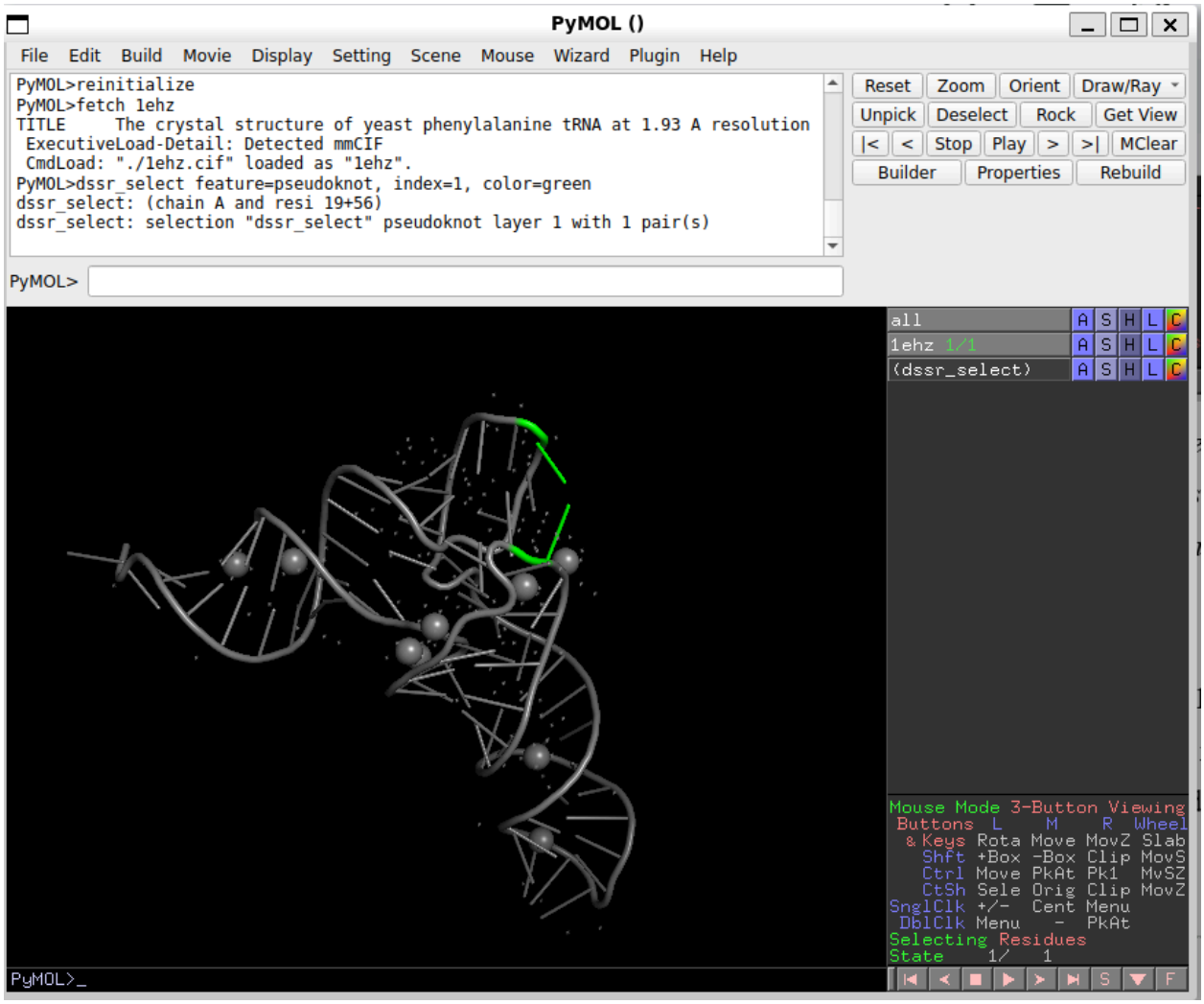
It can be seen in this image that the second instance of a hairpin in the yeast crystalline structure 1ehz is selected and colored in cyan. The object is created and named "my_hairpin" as shown on the right-hand side of the image. This use case demonstrates selecting a feature with the dssr_select command while using all possible parameters.

Sample Two: A more common usage of the dssr_select command. Although there are 10 adjustable parameters, some do not need to be changed and are sufficient as defaults in most common cases (precolor, show_info, quiet, exe). To run the command, it is not required that users use all parameters. Run the following to see a more simple use case:

1. reinitialize

2. fetch 1ehz
3. dssr_select feature=pseudoknot, index=1, color=green

Breakdown: First, reinitialize is called to make sure PyMOL is reset and ready to be used in case it was previously on another RNA structure. Then, using the 'fetch' command in PyMOL, the yeast crystalline structure is obtained by PyMOL and brought into the molecular viewer. Then, the dssr_select command is called such that the feature selected is a pseudoknot, and the index is set to 1 (meaning the first instance of a pseudoknot is to be selected), and it will be colored green. All the other parameters are set to their defaults automatically and run despite the user not typing them in. The following image is produced in PyMOL's molecular viewer:



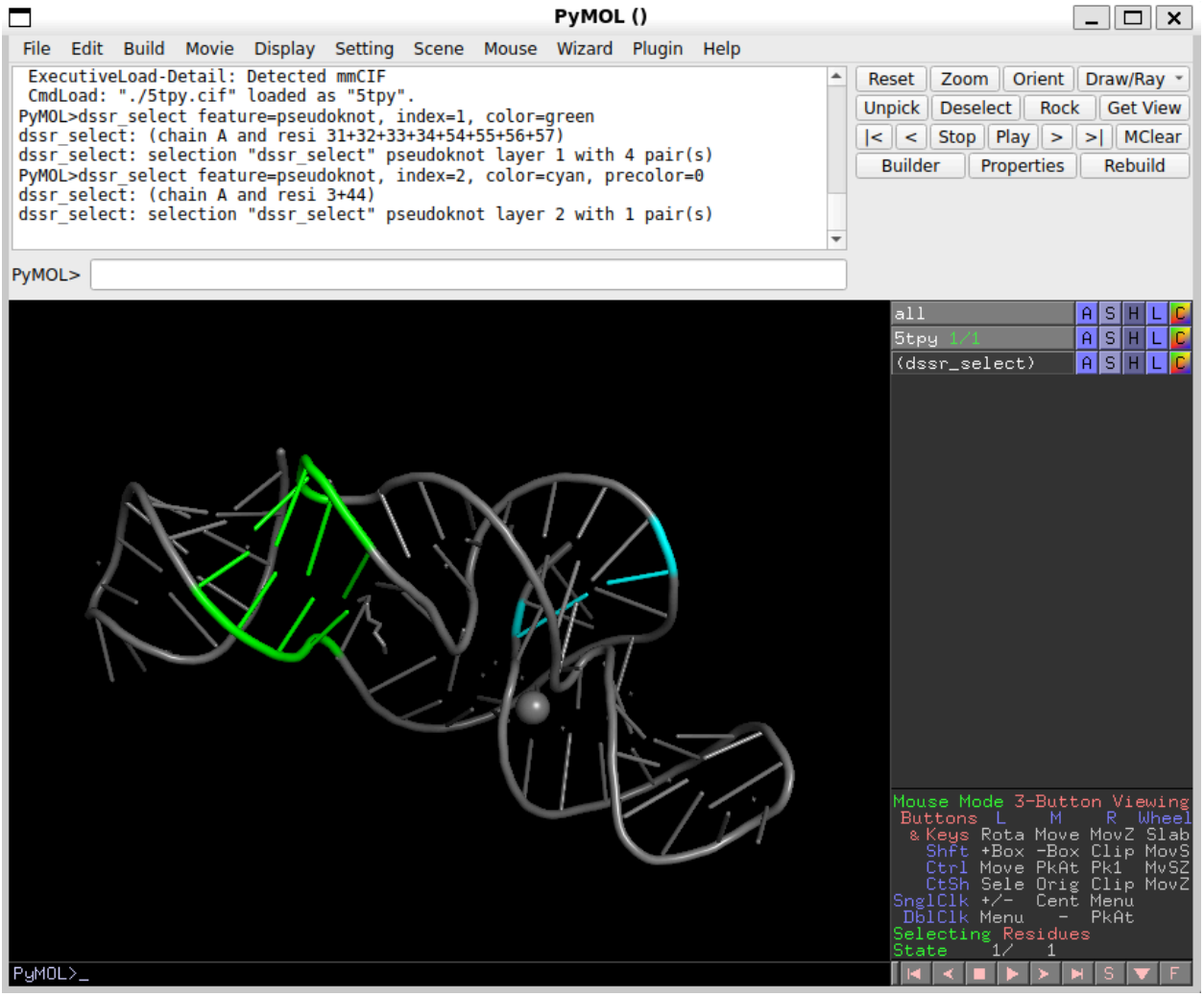
Here it can be seen that the first (and only) instance of a pseudoknot in the structure 1ehz was selected and colored green. On the right-hand side of the screen, the object created was named "dssr_select" because that is the default of the command. In addition, even though the user did

not manually type in precolor=1, the rest of the structure was colored to gray because it is a default of the dssr_select command.

Sample Three: Selecting multiple features from the CLI. The dssr_select command only allows users to select one feature at a time. However, through using it correctly and over multiple instances, the user can select multiple different features. Run the following commands to see a sample of this:

1. reinitialize
2. fetch 5tpy
3. dssr_select feature=pseudoknot, index=1, color=green
4. dssr_select feature=pseudoknot, index=2, color=cyan, precolor=0

Breakdown: First, reinitialize is called to make sure PyMOL resets before calling a feature. Then, the 'fetch' command is run in PyMOL to obtain the pdb structure 5tpy. Next, the dssr_select command is run to select the first layers of pseudoknots within the structure and color them to green with all other parameters being default. Next, the dssr_select command is run again to select the second layers of pseudoknots within the structure and color them cyan with precolor set to 0. Setting precolor to 0 will not make the entire structure gray before selecting, which means the previous selection of the first layers of pseudoknots are maintained. In other words, both the first layers and second layers of pseudoknots within the structure are able to be selected. The following image is produced from PyMOL's molecular viewer:



As seen in the image above, both the first and second layers of pseudoknots present in the pdb structure 5tpy are selected and visible in PyMOL's molecular viewer despite not being able to directly select two features at once with the command. This specific use case by taking advantage of the precolor parameter could be essential for selecting multiple of the same feature or of different types of features based on the user's needs.

Dssr_select allows for selection of the following features which can all be selected from the CLI and must be typed in this format:

1. aminors (A-minor motifs)
2. atom2bases (atom-base stacks)
3. bulges (bulge loops)
4. coaxstacks (coaxial stacks)
5. gquadruplexes (G-tetrads/G-quadruplexes)
6. pseudoknot (pseudoknot(s)
7. splayunits (splayed-apart units)
8. sssegments (single-stranded segments)
9. stacks (base stacking interactions)
10. stems (base-paired stems)
11. uturns (U-turns)
12. hairpins (hairpin loops)
13. helices (helical segments)
14. iloops (internal loops)
15. junctions (multi-helix junctions)
16. multiplets (base multiplets)
17. nonstack (non-stacking nucleotides)
18. nts (nucleotides)
19. pairs (base pairs)

Dssr_block: To create cartoon block-like schematics using DSSR in PyMOL's molecular viewer, the command 'dssr_block' can be run which has the following parameters:

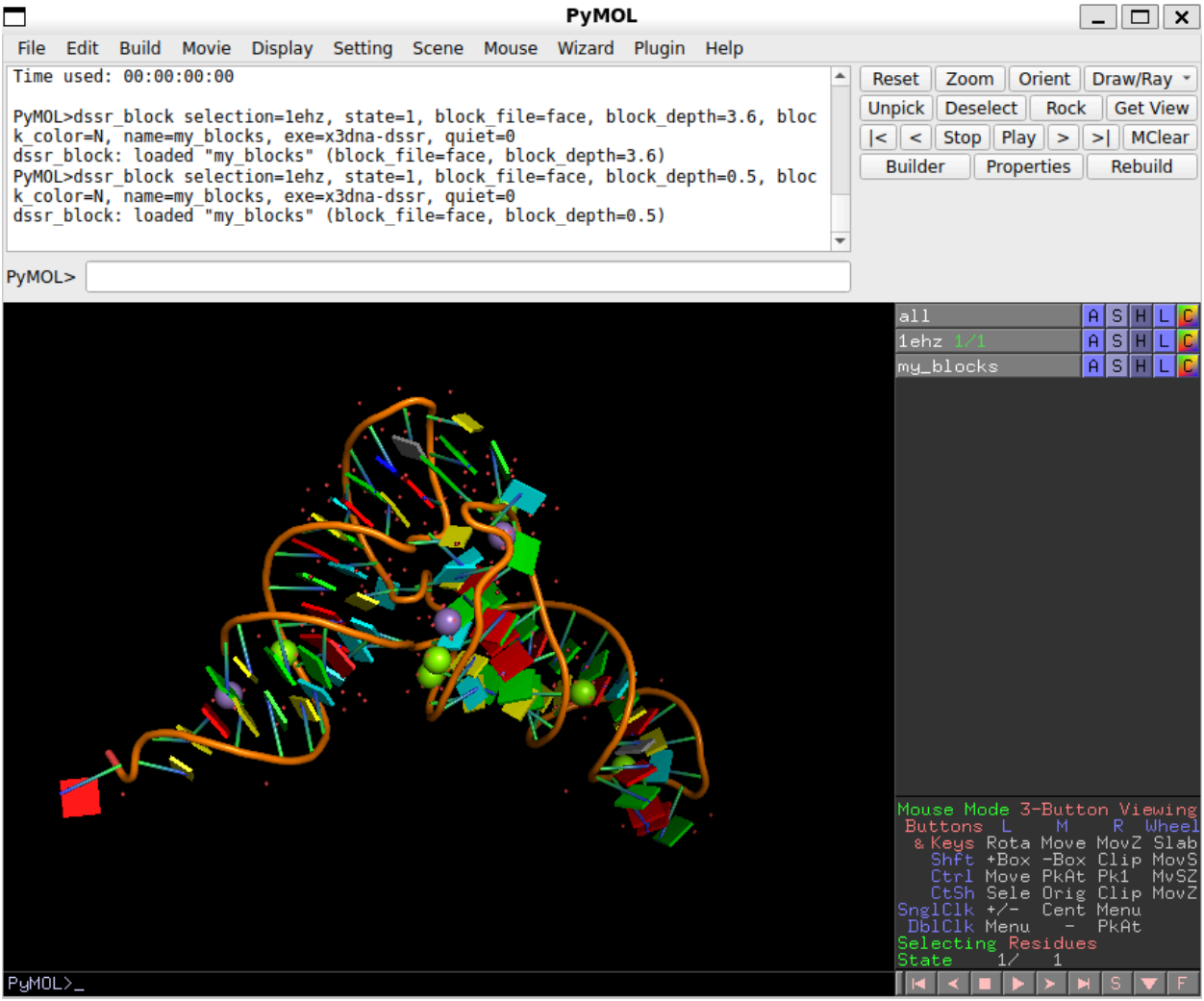
- selection: the atoms to make into block schematics
    - default: selection=all
- state: the state to use
    - default: state=-1
- block_file: the drawing style of the block
    - default: block_file=face
- block_depth: the visual thickness of the blocks
    - default: block_depth=0.5
- block_color: used to override the color of nucleic acid bases
    - default: (blank, only needed in rare cases)
- name: the name for the created object
    - default: name = (empty)
- exe: the path to the DSSR executable
    - default: exe=x3dna-dssr
- quiet: controls output
    - default: quiet=1

Sample One: Creating Cartoon Block Like Schematics for an entire structure can be done with the following command that uses all parameters of the command. Run the following commands to do so from PyMOL in the CLI:

1. reinitialize
2. fetch 1ehz
3. dssr_block selection=1ehz, state=1, block_file=face, block_depth=0.5, block_color=N, name=my_blocks, exe=x3dna-dssr, quiet=0

Breakdown: First, reinitialize is run to reset PyMOL to not let previous actions interfere with the current action. Then, using the 'fetch' command, pdb structure 1ehz is brought into the PyMOL molecular viewer. The dssr_block command is run, acting on the selection of 1ehz (the structure

that PyMOL has brought up), using the first state, with a default block depth of 0.5, default colors, named "my_blocks", using the default path to executable, and quiet set to 0. The following is generated by PyMOL:



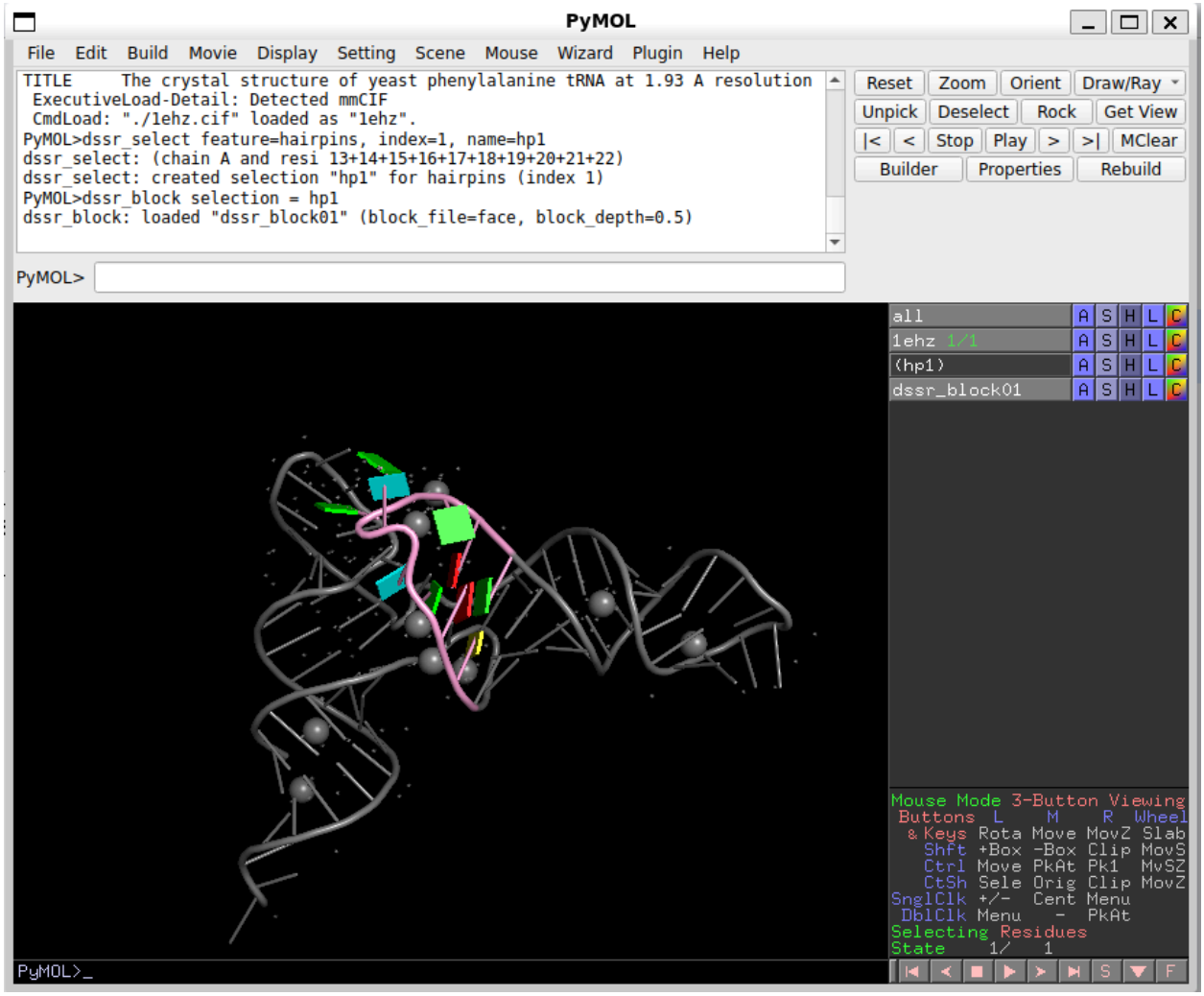
It can be seen that the entire structure of 1ehz is now composed of cartoon blocks to show the relationships between features within the structure. The created object "my_blocks" appears on the right-hand side of the screen and a message in the console tells the user that the dssr_block command was executed.

Sample Two: In some cases, it may be optimal to show the cartoon block schematics on certain parts of a structure rather than the entire structure. For example, a user may want to see the

planarity between a pseudoknot relationship in one example. To run a sample of creating block-like schematics on a hairpin loop, run the following commands:

1. reinitialize
2. fetch 1ehz
3. dssr_select feature=hairpins, index=1, name = hp1
4. dssr_block selection=hp1

Breakdown: First, reinitialize is run to reset PyMOL to not let previous actions interfere with the current action. Then, using the 'fetch' command, pdb structure 1ehz is brought into the PyMOL molecular viewer. The dssr_select command is used to select the first instance of a hairpin loop and name it as 'hp1'. Next, the dssr_block command is run on the 'hp1' object as a means of only creating cartoon block schematics for the generated selection. The following image produced by PyMOL is shown below:

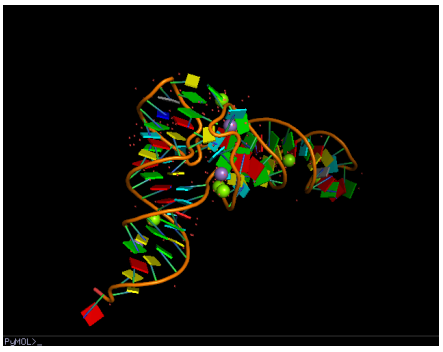


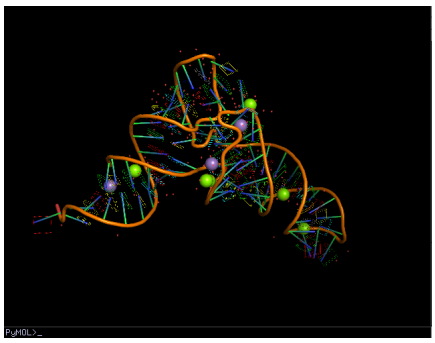
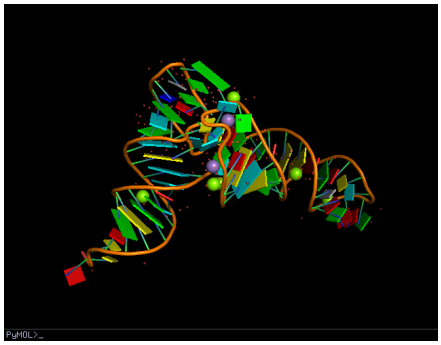
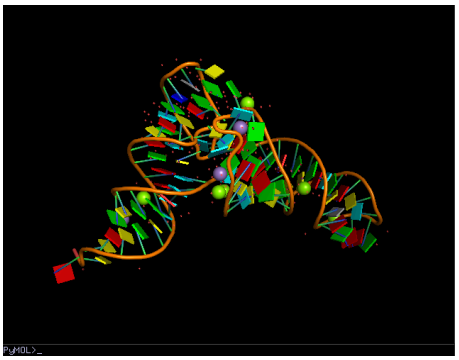
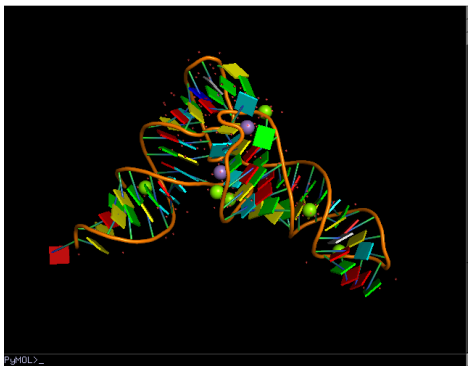
In this image, it can be seen that a hairpin loop was selected and is colored in pink as a result of the dssr_select command. In addition to this, there are cartoon blocks representing the interactions of nucleotides within the hairpin loop.

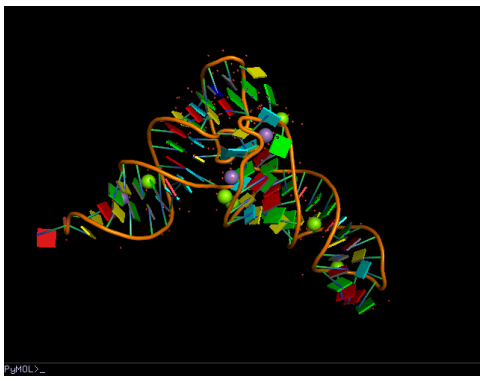
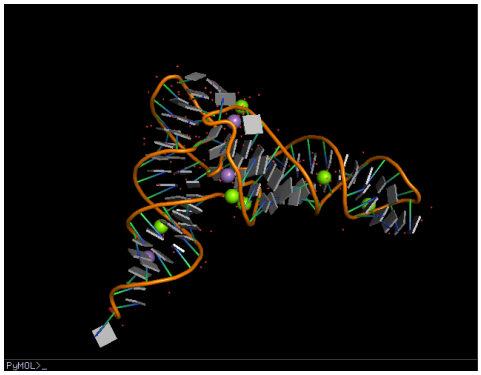
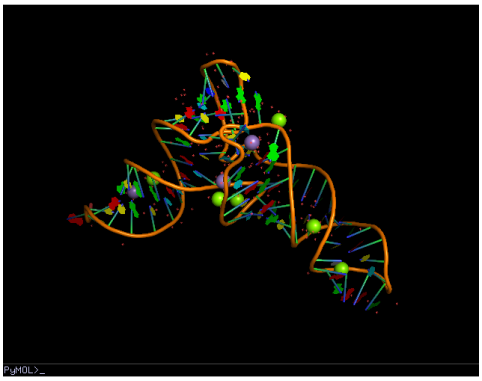
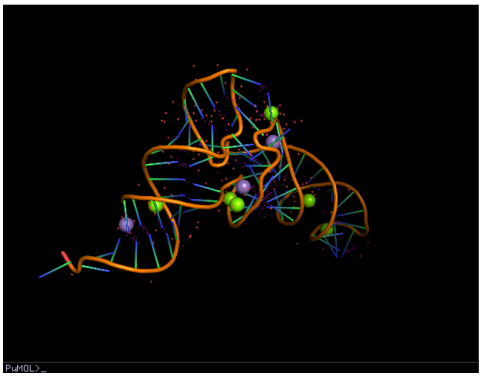
Dssr_block allows for 9 different types of inputs for the block_file command as supported styles:

- "face"
- "edge"
- "wc"
- "g4"
- "equal"
- "minor"
- "gray"
- "fill"
- "hbond"

Here is a table showing the visual representation of each block_style used on pdb structure 1ehz:

Block_file input	Image
face	

edge	
wc	
g4	
equal	

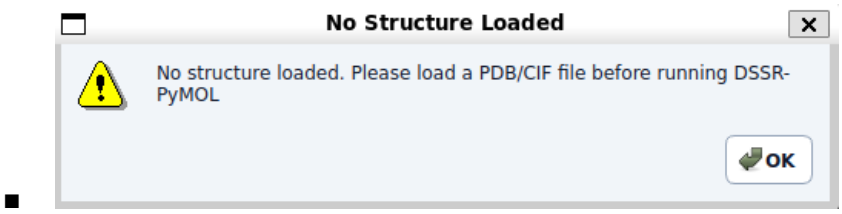
minor	
gray	
fill	
hbond	

Dssr_select from the GUI (Graphical User Interface):

One of the most effective ways to use the Dssr_select software is from its GUI. The Graphical User Interface provides users with multiple ways of selecting features, significant customization through selection, and ease of use with buttons rather than having to type in commands. There are three ways to access the GUI of Dssr_select:

1. In PyMOL, run the following command in the CLI:
    a. dssr_2d
2. In PyMOL, run the following command in the CLI:
    a. dssr_gui
3. In PyMOL, click plugin, then click DSSR

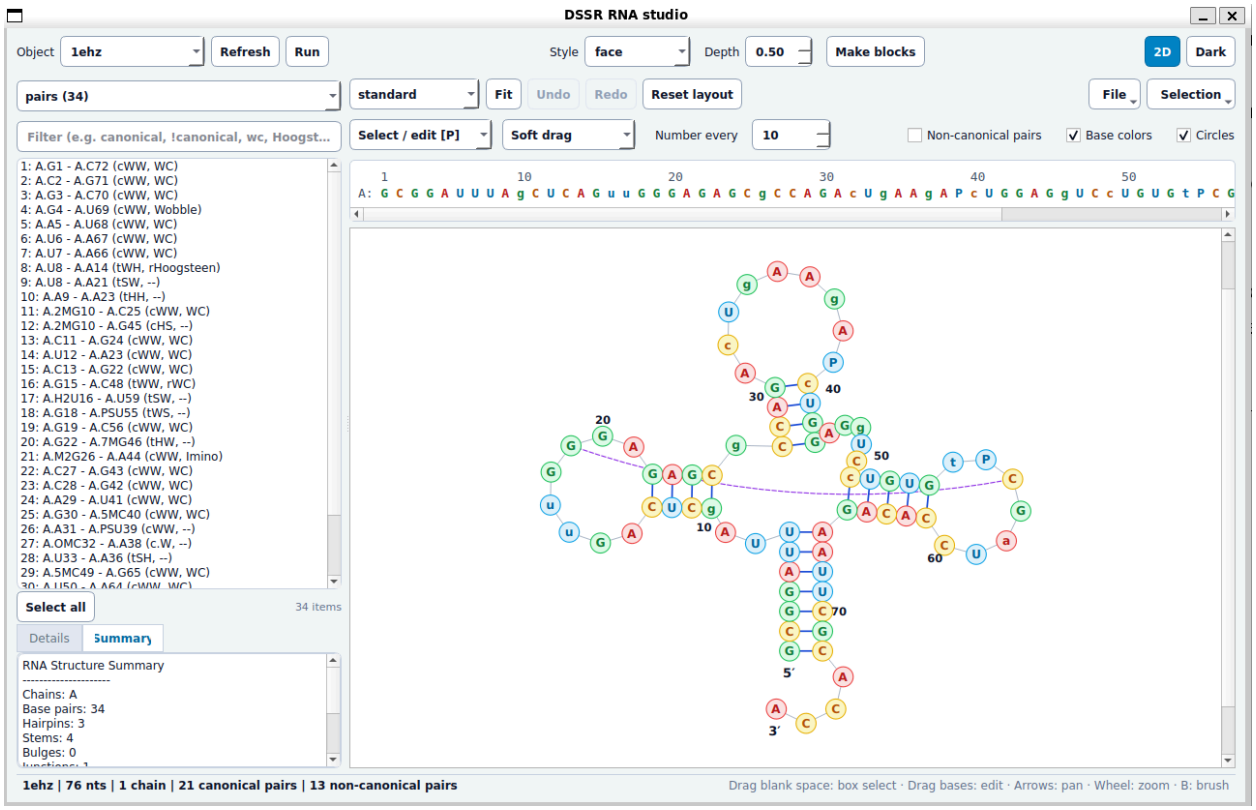
NOTE: A pdb structure should be loaded or fetched into PyMOL before running any of the commands or clicking buttons to access the GUI. If the user does not, they will be prompted with a message to enter a feature first before they can use the GUI.



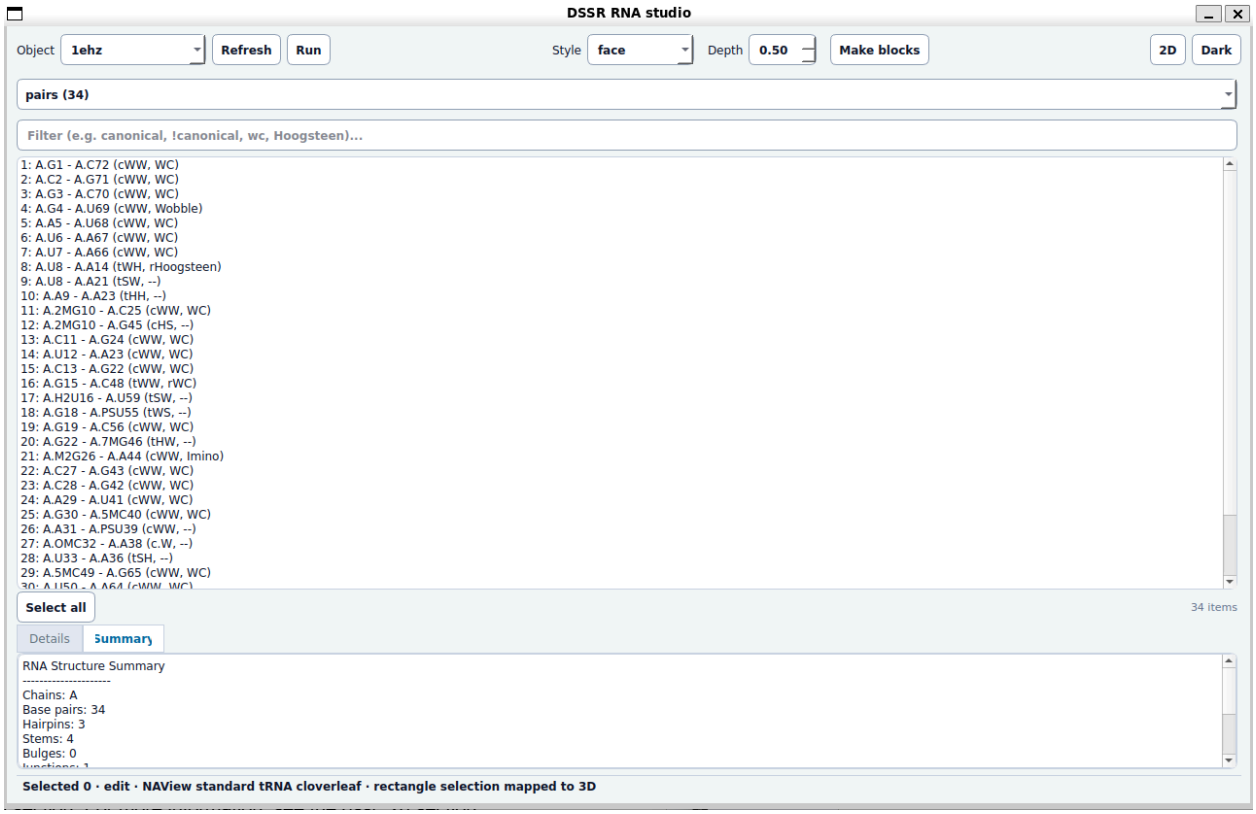
Sample One: Launching the Dssr_select GUI, run the following commands:

1. fetch 1ehz
2. dssr_gui

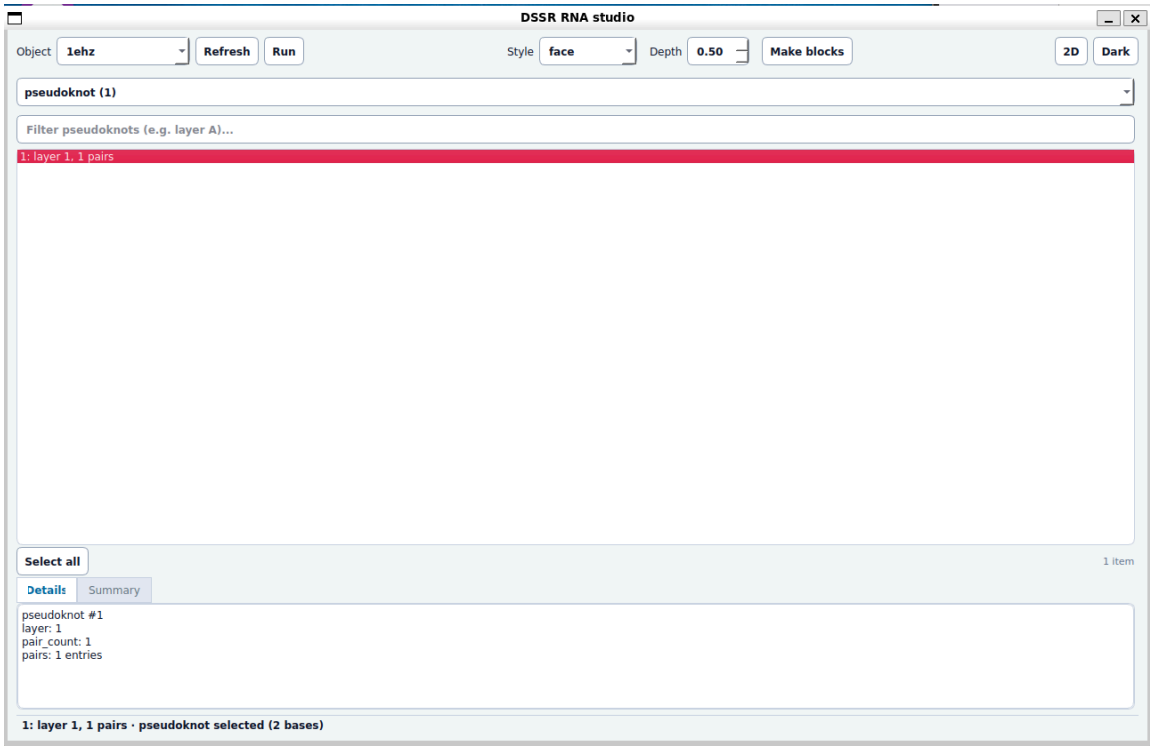
The following GUI Window will pop up on the user's screen after running the commands:



Note that in the top right corner, a button labeled with "2D" is highlighted in dark blue. This indicates that it is currently selected. By clicking on this button, 2D mode can be disabled and the 2D diagram of 1ehz will disappear. Here is a picture of the GUI after deselecting the 2D button:



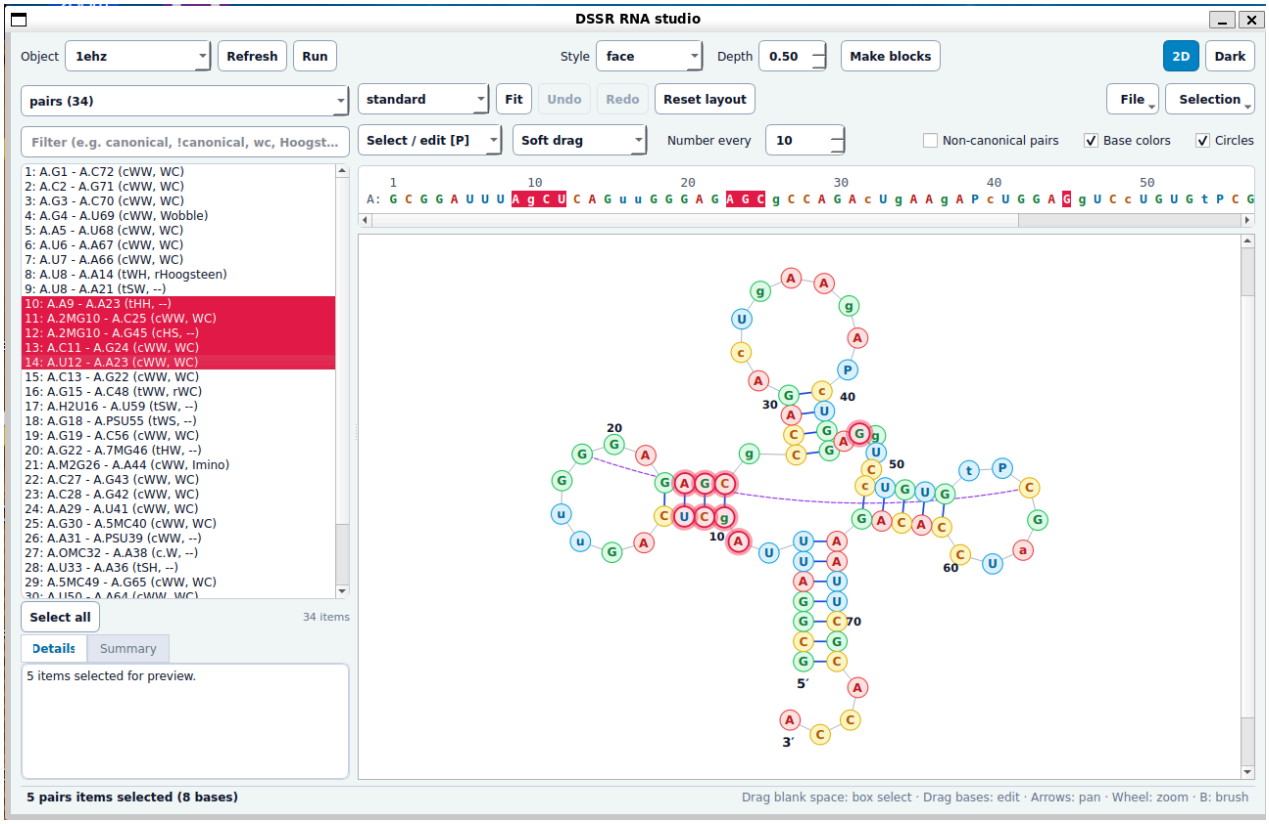
Sample One: To see the available features on a given structure, click on the drop-down menu at the top of the page. By default, the menu should contain “pairs” and a number adjacent to it in parentheses like (34). This indicates that the current list shown in the GUI from 1-34 are all instances of pairs. Clicking on the drop down menu, will show the list of all the features and the number of occurrences of that feature for the given structure (in this case 1ehz). For example, to select a pseudoknot, click on the drop-down menu and select pseudoknots. Then, click the only instance of a pseudoknot in 1ehz. The following will be shown in the GUI and in PyMOL:



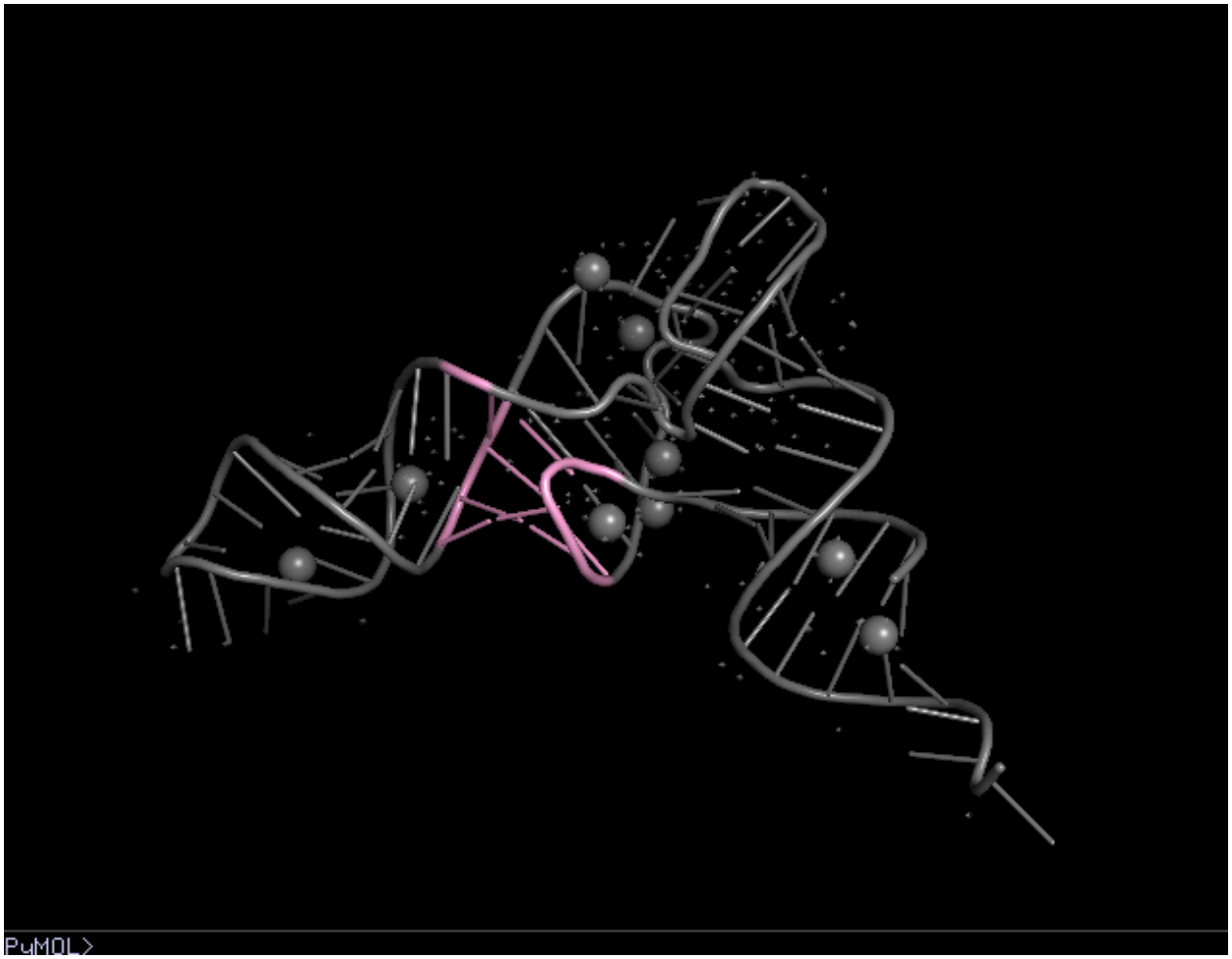
The red highlight indicates that the current instance of the feature has been selected. The selection is mirrored in PyMOL's 3D molecular viewer as well.



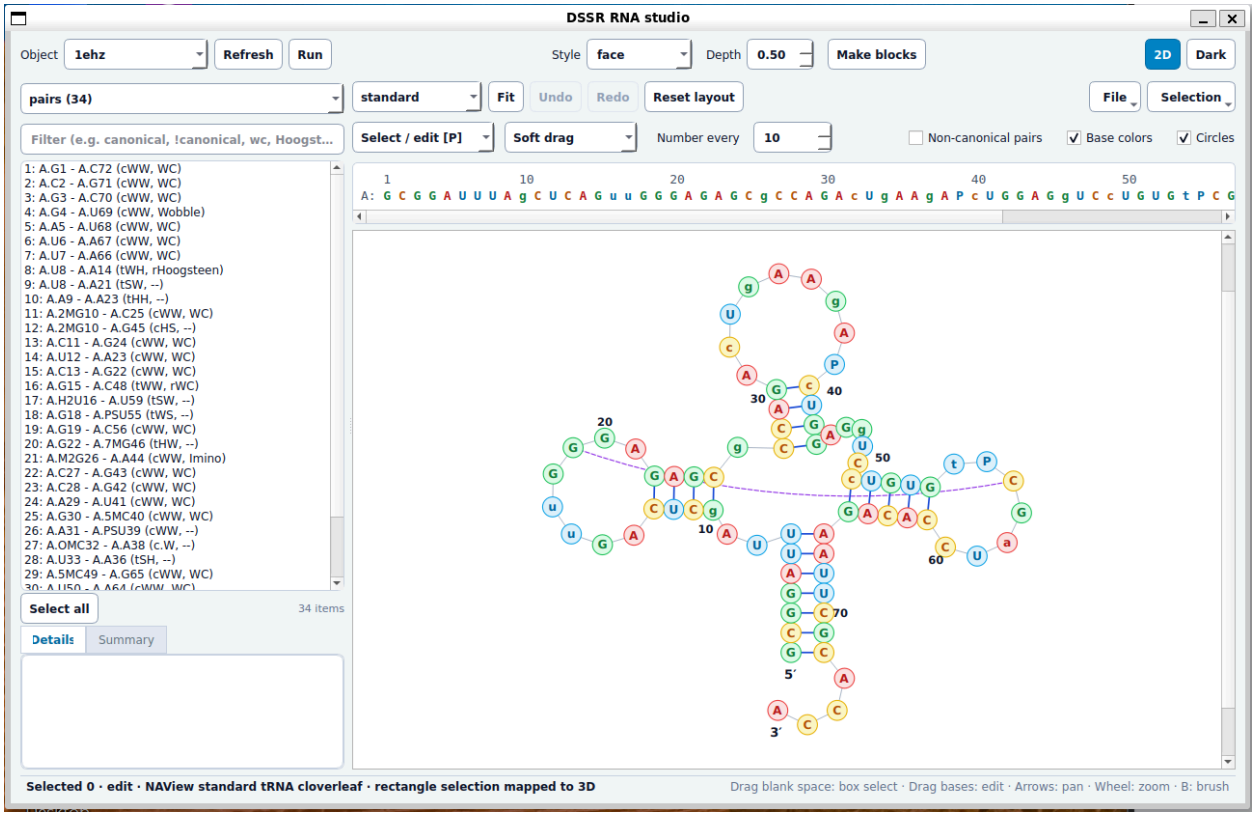
Sample Two: Selecting multiple features at once is made simple through the GUI. In this example, the selection of multiple features will be shown in both the 2D and the 3D representation of 1ehz. To do this, ensure the 2D button in the upper right corner of the screen is toggled. Then, holding the shift button on the keyboard, double click the indices of pairs from 10 to 14 to produce the following output:



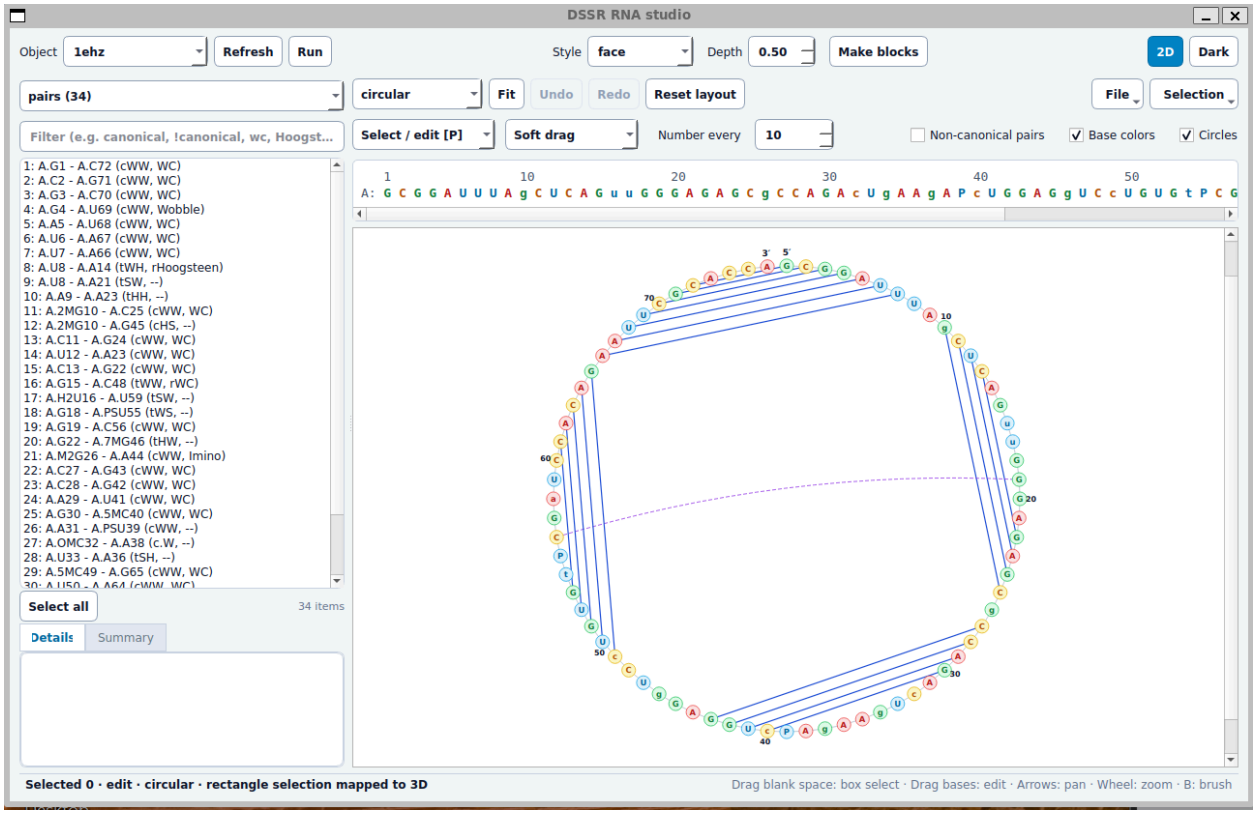
The circles highlighted in red on the 2D representation of 1ehz are intended to match the highlighted structures shown on the left-hand side of the screen. Note that for the selections to be created in PyMOL's molecular viewer, the user must double click on their selections in the GUI which will get rid of the highlights on the 2D model shown above.



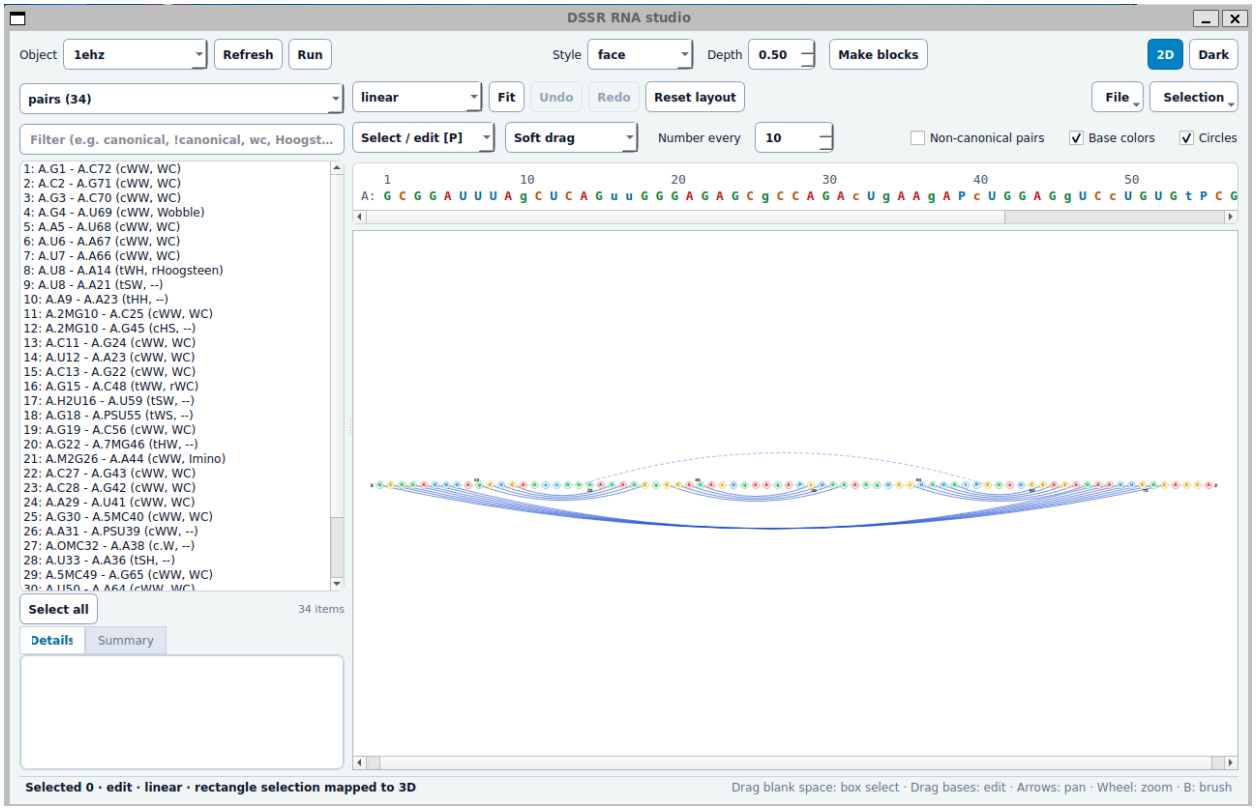
Samples Three-Six: Different 2D model representations of Nucleic Acid structures in the GUI are available. The GUI offers four different options which can be toggled by a drop-down menu located in the upper left side of the screen that is labeled as ‘standard’ by default. It has four options for models: ‘standard’, ‘circular’, ‘linear’, and ‘radiate’. Shown below, are the different models:



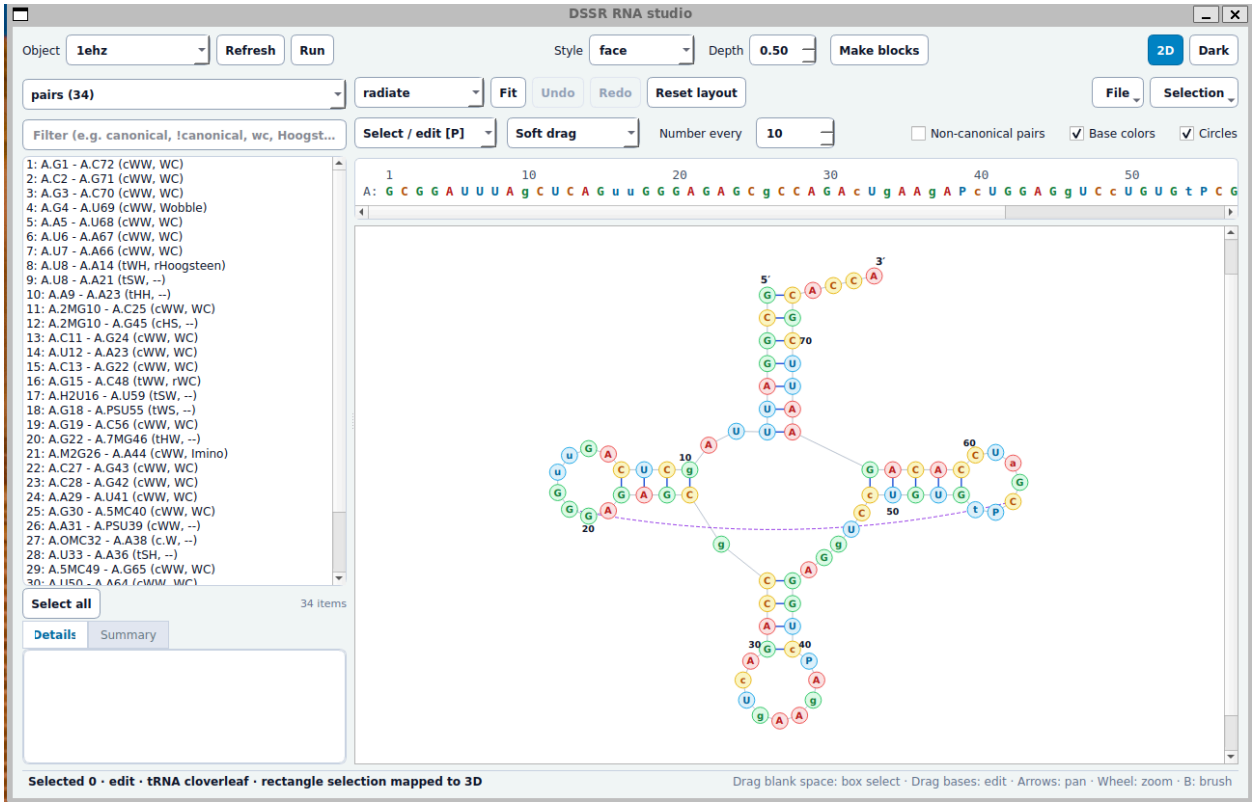
This represents the 2D model for 1ehz with no selections made on the 'standard' model. Note that the drop-down menu is set to standard in the upper left corner of the screen.



This represents the 2D Model for 1ehz with no selections made on the 'circular' model. Note that the drop-down menu is set to circular in the upper left corner of the screen.

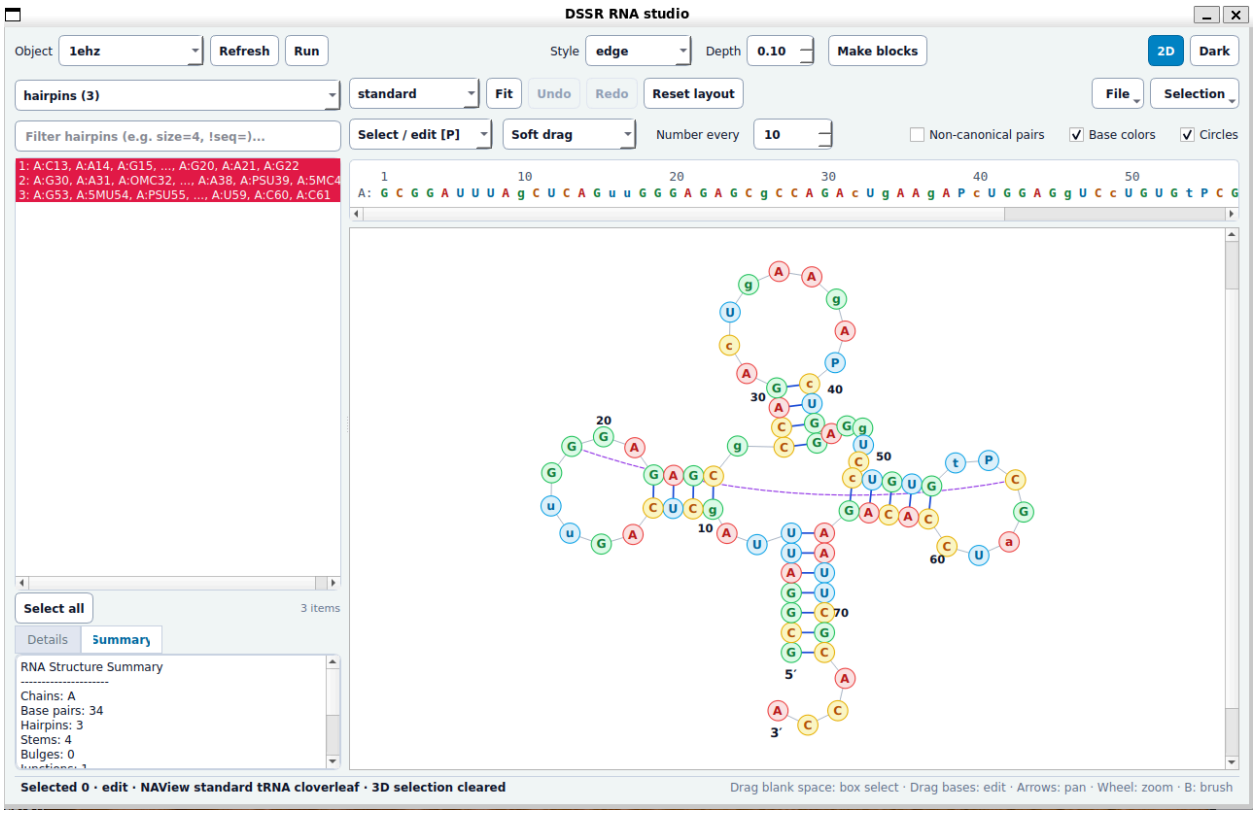


This represents the 2D Model for 1ehz with no selections made on the 'linear' model. Note that the drop-down menu is set to linear in the upper left corner of the screen.

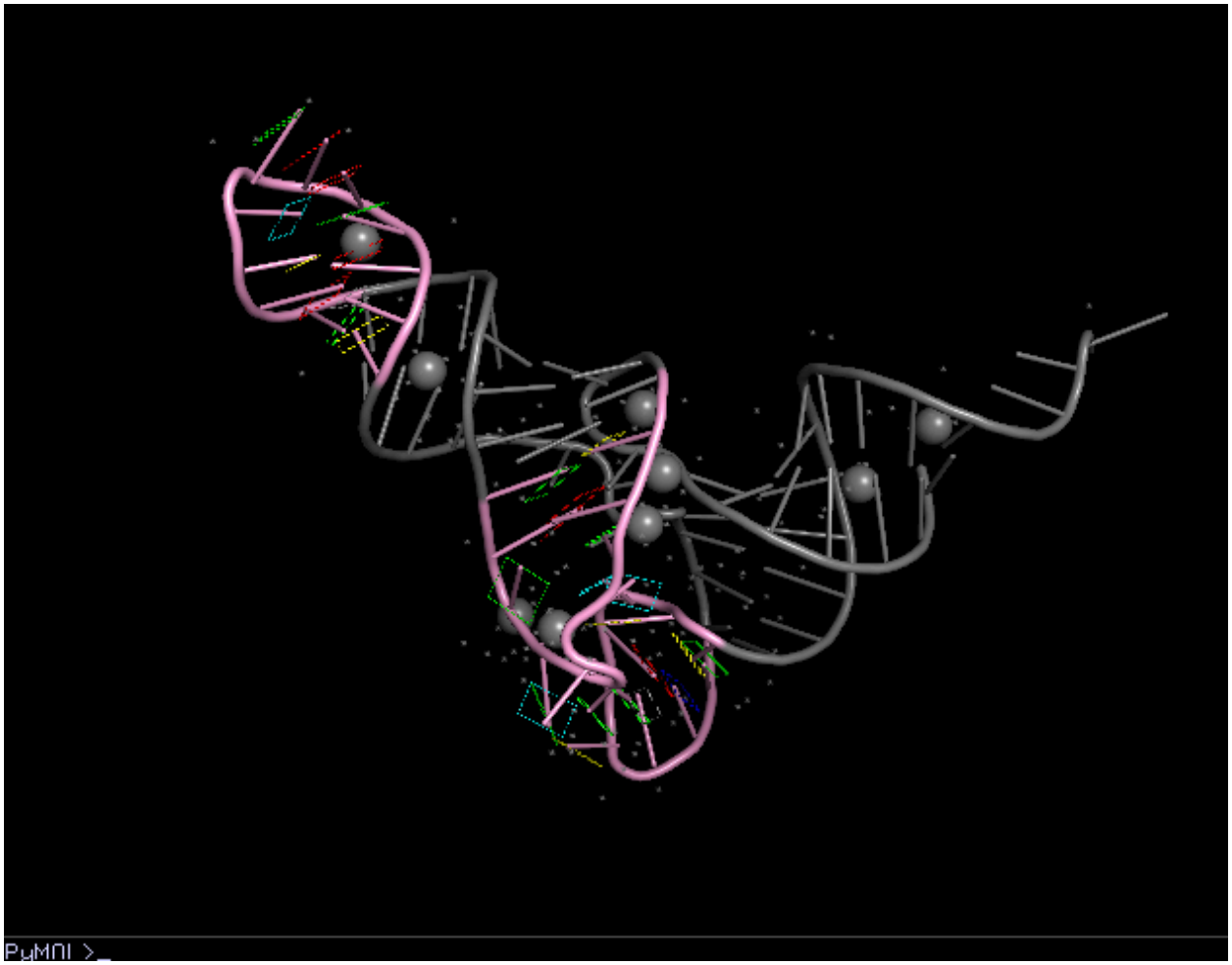


This represents the 2D Model for 1ehz with no selections made on the 'radiate' model. Note that the drop-down menu is set to radiate in the upper left corner of the screen.

Sample Seven: Creating cartoon block-like schematics is made simple from the GUI. First, select a feature or features by double-clicking on them in the left-hand side of the screen and then press the "Make blocks" button centered in the top of the screen to create blocks on the selected features. To make the whole structure out of block-like schematics, ensure no selections are made and then press "Make blocks." The style and depth of the blocks are able to be edited by the drop-down menu labeled "Style" and the box labeled "Depth" that allows the user to type in an appropriate value (0-3.6). In this example, every hairpin loop for 1ehz is selected and blocks are made with a depth of 0.10 using the 'edge' style:

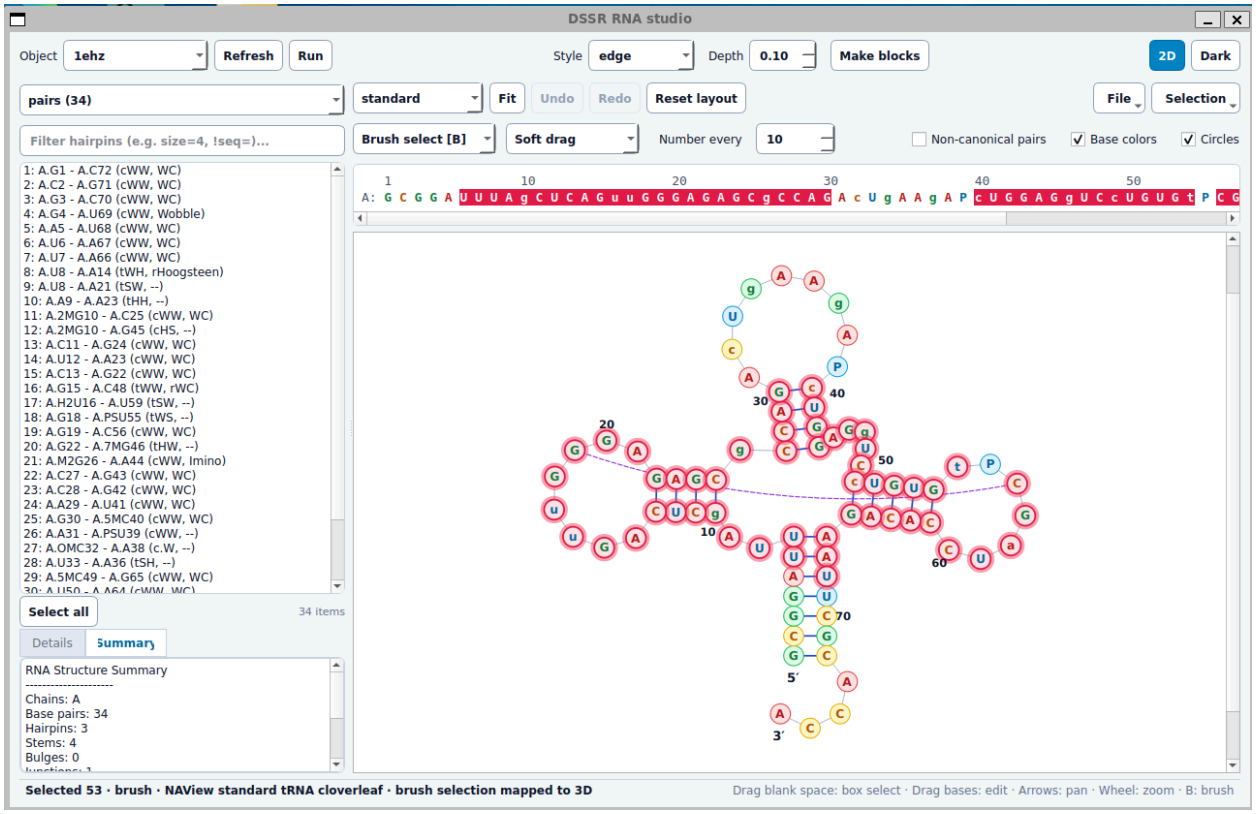


This image shows the setup for making the cartoon block schematics for every hairpin loop in the 1ehz structure. Below, is the created image in PyMOL's molecular viewer:

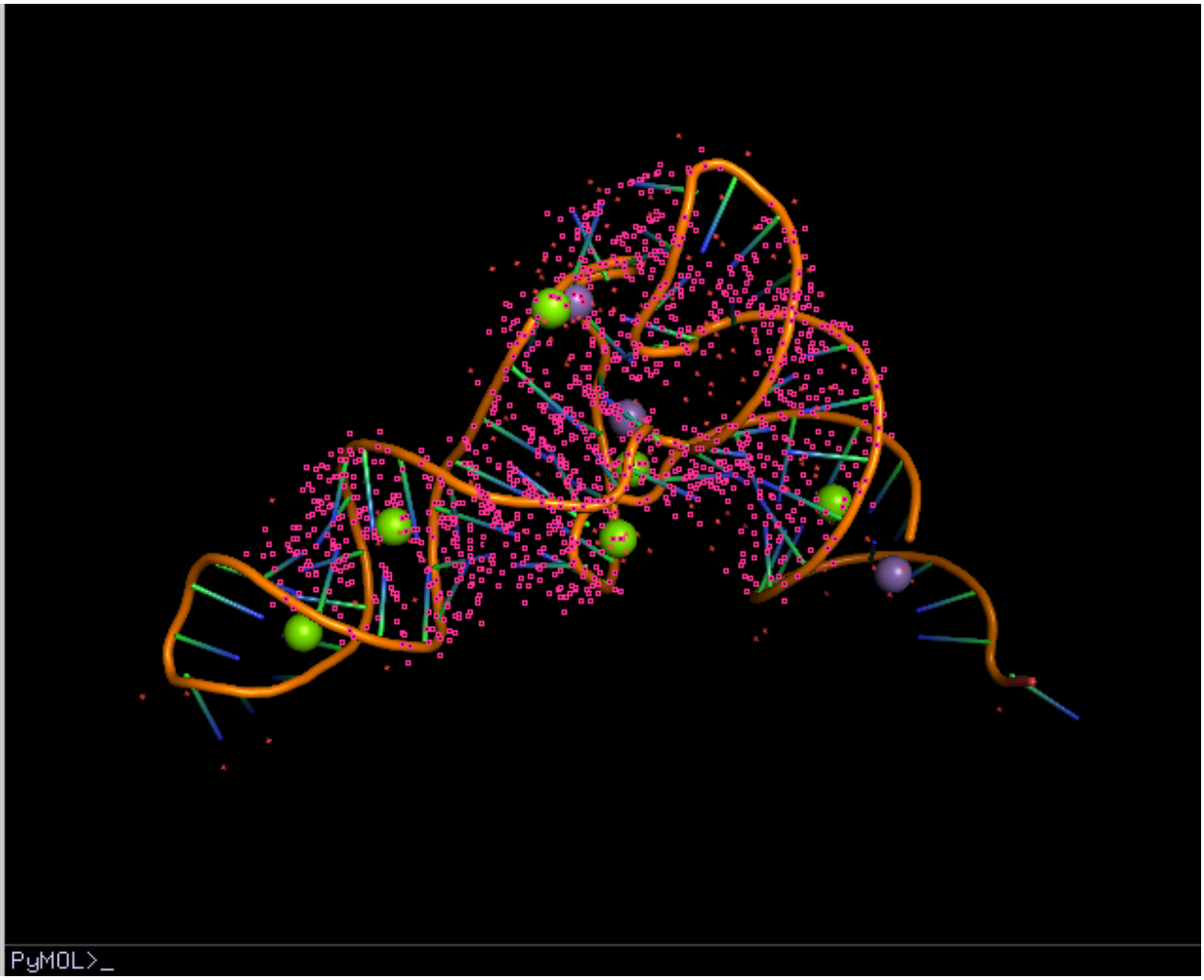


This image shows the created cartoon block schematics for every hairpin loop with the edge style and a depth of 0.10 as well as all of the hairpin loops selected and highlighted in pink for the structure 1ehz.

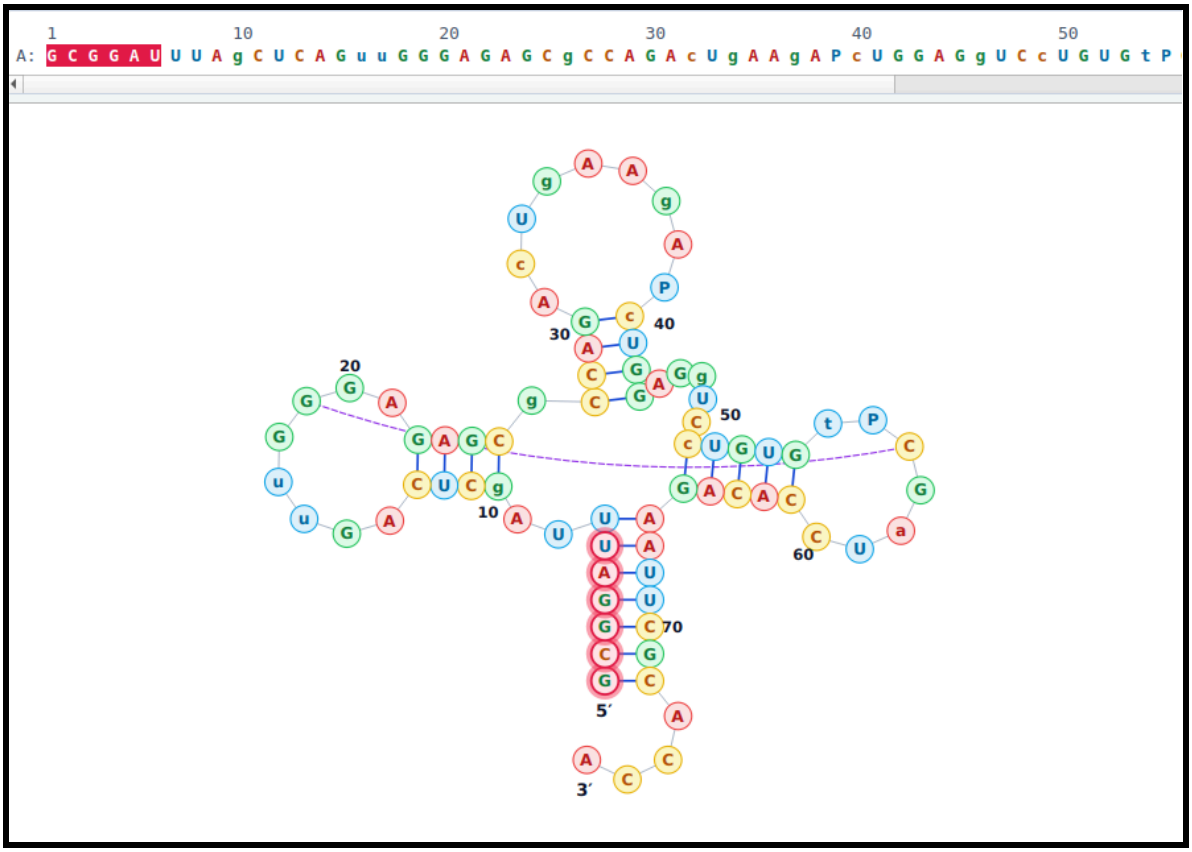
Sample Eight: To select individual nucleotides using the GUI, there are two modes toggled with the keyboard keys ‘p’ and ‘b’. Clicking ‘p’ goes on select/edit mode where the user can directly click on a nucleotide on the 2D diagram and the atoms will be highlighted or shown by PyMOL in its molecular viewer. Clicking ‘b’ goes on brush select mode where the user can drag their cursor across multiple nucleotides on the 2D diagram and the atoms will be highlighted or shown by PyMOL in its molecular viewer.



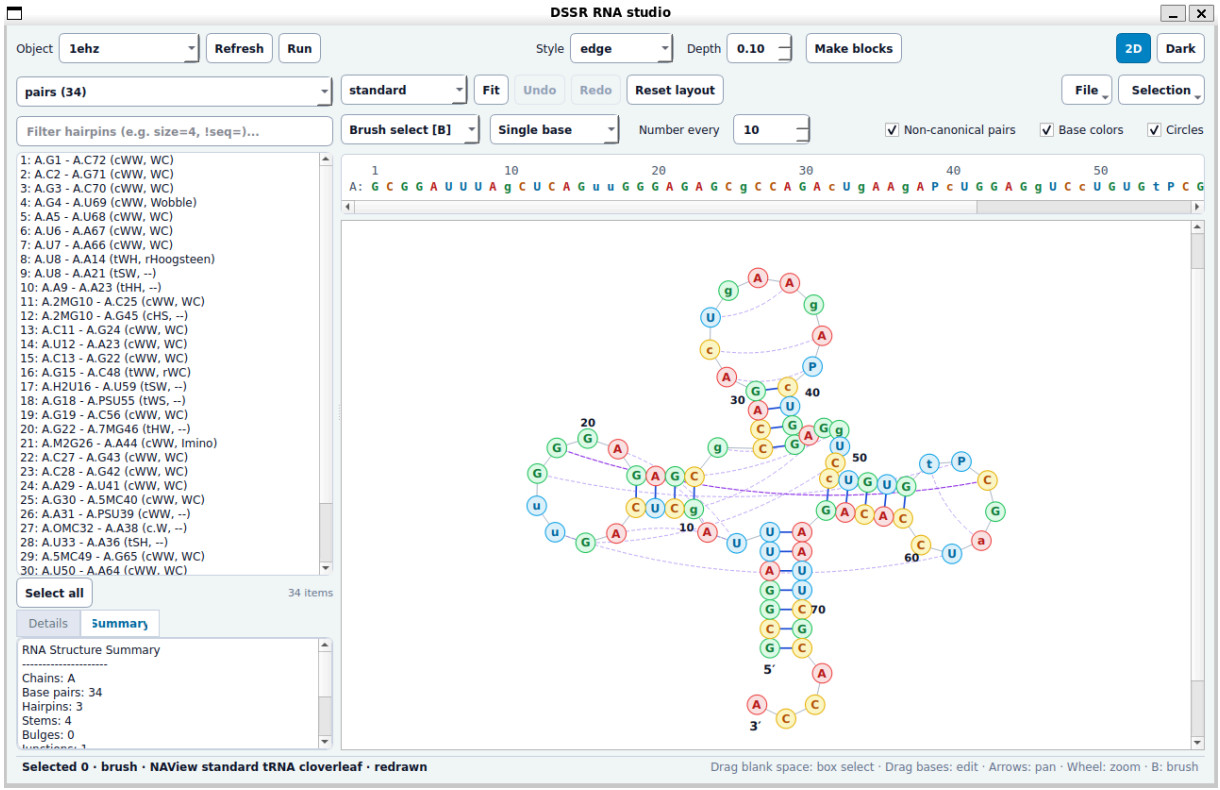
This image shows nucleotides selected using the brush select mode and dragging the cursor across the screen to select the highlighted nucleotides. Shown below, is PyMOL's atomic highlighting of the nucleotides shown above in the 2D viewer.



Note that the atoms involving the nucleotides can be individually selected the same way by clicking on the ruler above the 2D visual model of the structure. An example is shown below:

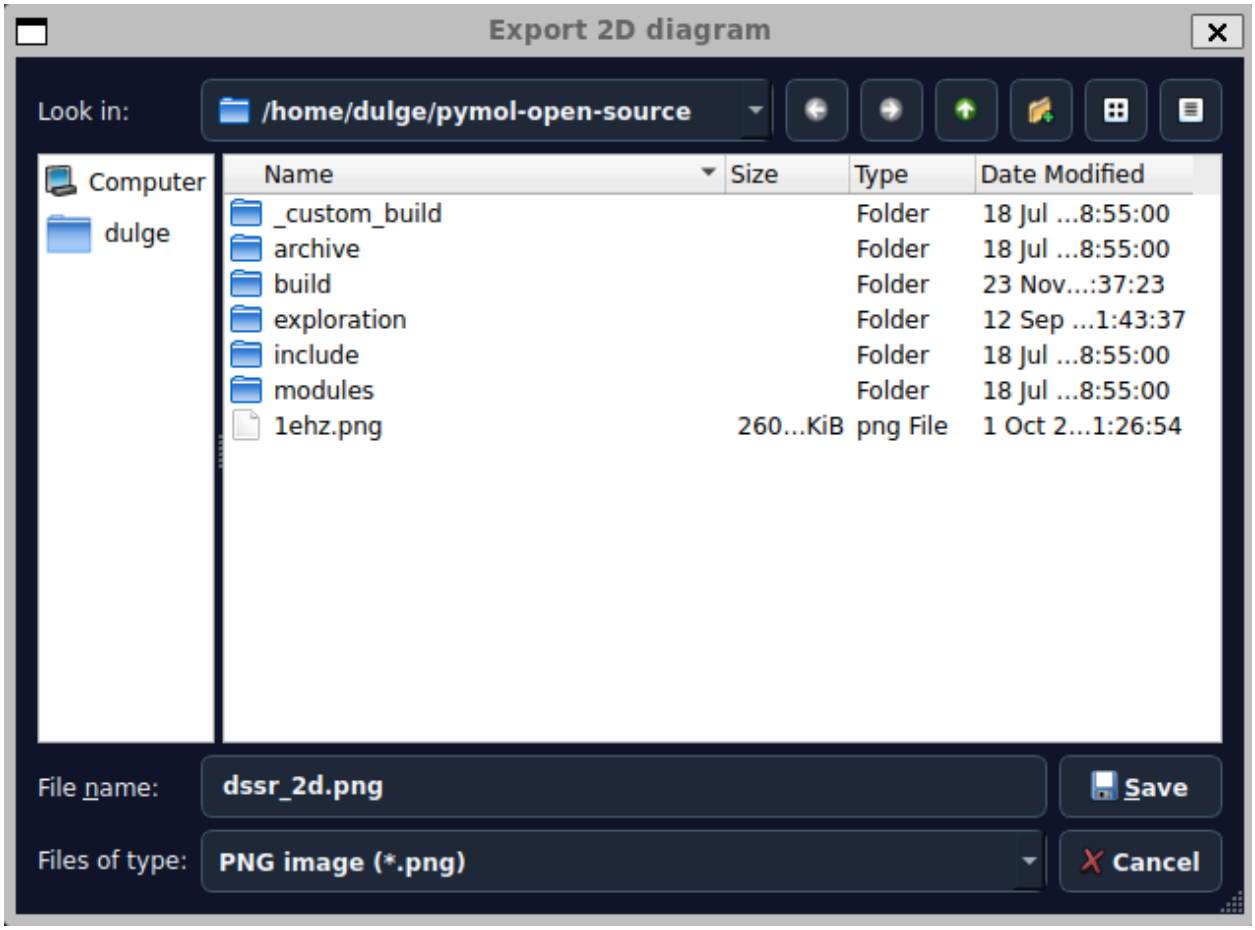


Sample Nine: To show non-canonical pairs in the 2D model of a given structure (in this case 1ehz), check the box labeled “Non-canonical pairs”. As a result, dashed lines will appear between nucleotides to show the non-canonical relationships that exist.



Note that in a previous example showing the 2D model of the structure 1ehz in this model, there are no dashed lines representing the non-canonical pairs between nucleotides. Also note that the buttons adjacent to "Non-canonical pairs" can be toggled to affect the 2D model visually. For instance, unchecking "Base Colors" will make the color of every nucleotide shown grey and unchecking "Circles" will get rid of the circular structure that each nucleotide has and simply replace it with a letter.

Sample Ten: Exporting an image of the 2D model is made simple in the GUI. Click the file button and click export image to be prompted with the following image shown below. It is available to save the image as a .png or as a .svg file for user ease.



Note, when the user clicks file, other options are available. For example, the user can save the layout as a .json file, load a layout that has already been saved, or copy the sequence or DBN (Dot Bracket Notation) to clipboard.

Sample Eleven: Loading a different nucleic acid structure into PyMOL will not automatically update the GUI. To do so, first click the "Refresh" button, then click on the object drop-down menu and select the new loaded structure, followed by the "Run" button in the upper left corner. This should update the GUI and the 2D model accordingly.

Note: There are other buttons in the GUI that allow for simple modifications visually and aesthetically and should be used at the user's discretion but are not necessary in most cases such as fit (adjusting the 2D model), Number Every (numbering for the ruler above the 2D model), and Dark (makes the screen go on dark mode). There is also the Select All button that will select every instance of a feature. Below this, the user can toggle between Details and Summary such

that Summary will provide information about the provided RNA structure. These features are not given their own samples and are self explanatory for the most part.